



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

AIOT

10-07-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	A	A	C	A	C	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	B	C	C	A	A	B	B	A
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1000	2	5	24	20	1	24	8	20	5

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	C	D	C	D	C	C	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	D	B	B	C	C	C	C	D
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	2	5	10	6	8	1	11	6	3

PART-3 : MATHEMATICS

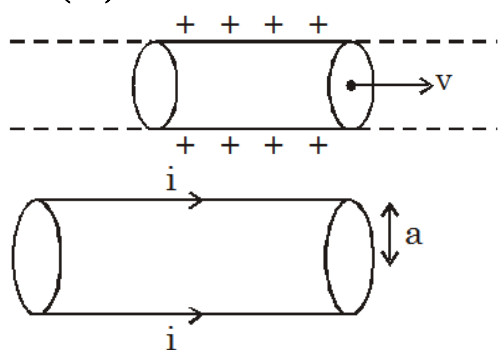
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	C	D	A	C	D	A	D	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	C	B	C	A	D	C	C	C
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	0	0	1	20	4	624	11	2	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)



$$E = \frac{2k\lambda}{a}$$

$$E = 2 \frac{1}{4\pi\epsilon_0} \frac{\lambda}{a}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 a}$$

$$\lambda = 2\pi\epsilon_0 a E$$

$$\lambda = 2\pi\epsilon_0 a E$$

$$q = 2\pi\epsilon_0 a E x$$

$$i = \frac{dq}{dt}$$

$$i = 2\pi\epsilon_0 q E v$$

2. Ans (B)

Radius of the new sphere

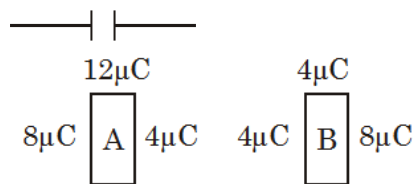
$$\frac{4}{3}\pi r^3 = 8 \times \frac{4\pi}{3} \left(\frac{10^{-3}}{2}\right)^3, r = 1 \text{ mm}$$

$$\text{Total charge} = 8 \times 5 \times 10^{-9} = 40 \times 10^{-9} \text{ C}$$

$$\text{Capacitance} = C = 4\pi\epsilon_0 r$$

$$U = \frac{Q^2}{2C} = 7.2 \text{ mJ}$$

3. **Ans (C)**



$$RC = 1 \times 2$$

$$\theta_{\text{right}} = \theta_0 e^{-\frac{t}{RC}}$$

$$\theta_{\text{right}} = 4e^{-\frac{t}{2}} \mu\text{C}$$

$$\theta_{\text{net}} = 8 + 4e^{-\frac{t}{2}} \mu\text{C}$$

4. **Ans (A)**

$$\frac{\phi}{t} = \frac{EA}{t} = \frac{Nm^2}{CbT}$$

$$= \frac{M^1 L^1 T^{-2} L^2}{AT T}$$

$$\frac{\phi}{t} = [M^1 L^3 T^{-4} A^{-1}]$$

5. **Ans (A)**

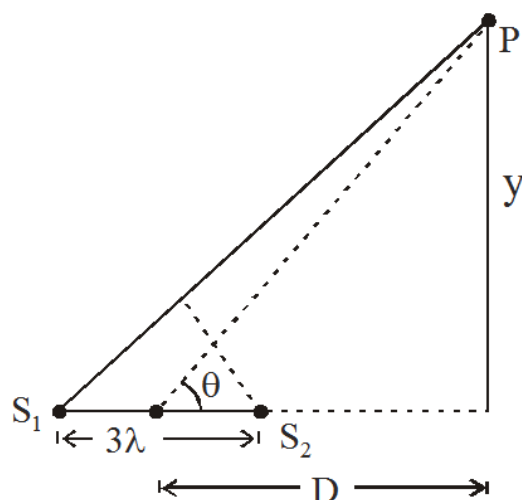
In both the cases magnetic field at O will be zero.

7. **Ans (A)**

$$h_{\text{app}} = \frac{h_{\text{real}}}{\mu}$$

$$\therefore \int_0^{10} \frac{dz}{1.63 \left(1 + \frac{z}{20}\right)} = \frac{20}{1.63} \ln \frac{3}{2} = 5 \text{ cm}$$

8. **Ans (C)**



$$S_1P - S_2P = 3\lambda \cos \theta = 2\lambda$$

$$\Rightarrow \cos \theta = \frac{2}{3} = \frac{D}{\sqrt{D^2 + y^2}} \Rightarrow 5D^2 = 4y^2$$

$$\therefore y = \frac{\sqrt{5}}{2} D$$

9. **Ans (A)**

$$A_t = \frac{A_0}{2^{10}} = \frac{A_0}{1024}$$

$$\therefore \text{Prince} = \frac{4096}{1024} = 4 \text{ Rs}$$

10. **Ans (C)**

$$d = \sqrt{2Rh}$$

$$\therefore \frac{d}{d'} = \sqrt{\frac{h}{h'}}$$

$$\Rightarrow 2 = \sqrt{\frac{h}{h'}} \Rightarrow h' = 4h = 4 \times 160 \text{ m} = 640 \text{ m}$$

$$\therefore \text{Increase in height} = 640 - 160 = 480 \text{ m}$$

11. **Ans (C)**

$$\text{Separation} = f_0 + 4f + f_e = 180 + 4(3.5) + 5 = 199 \text{ cm}$$

12. **Ans (C)**

At closest approach there velocity will be equal

$$\text{to } v. m_p \cdot u = m_p v + 4 m_p \cdot v$$

$$\Rightarrow v = \frac{4}{5} u$$

$$\frac{1}{2} m_p u^2 = \frac{1}{2} m_p \left(\frac{4}{5} u\right)^2 + \frac{1}{2} 4 m_p \left(\frac{4}{5} v\right)^2 + \frac{K \cdot 2e^2}{r}$$

$$\Rightarrow r = \frac{5K e^2}{m_p \cdot u^2}$$

13. **Ans (B)**

B, as temperature increase

14. **Ans (C)**

$$2mg \sin \theta \leq mg (\mu_1 \cos \theta + \mu_2 \cos \theta)$$

$$2 \tan \theta \leq \mu_1 + \mu_2$$

$$\tan \theta_{\text{max}} = \frac{\mu_1 + \mu_2}{2}$$

15. **Ans (C)**

Apply Bernoulli's equation at the required points

16. **Ans (A)**

For the system to be bounded one, total energy of the system must be negative. So, object having total energy E_1 bounded and other is unbounded.

17. **Ans (A)**

Initially increase in temperature increases the density and after $T = 4^\circ\text{C}$ increase in temperature decreases the density.

18. **Ans (B)**

$$\therefore f' = \left(\frac{V_{\text{sound}} \pm V_0 + V_{\text{wind}}}{V_{\text{sound}} \mp V_{\text{source}} + V_{\text{wind}}} \right) f$$

$$\therefore f' = f$$

19. **Ans (B)**

apply work-energy theorem

$$mgz - \left[\frac{1}{2} K \left(\frac{mg}{K} + z \right)^2 - \frac{1}{2} K \left(\frac{mg}{K} \right)^2 \right] + Fz = 0$$

$$\Rightarrow z = 2F/K.$$

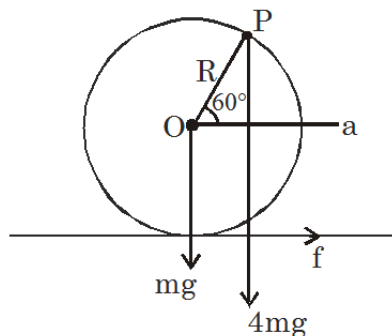
20. **Ans (A)**

$$\tau_0 = I_0 \alpha$$

$$\Rightarrow \frac{2}{5} mR\alpha + f = 2mg$$

$$\& f = ma = m\alpha R$$

$$\Rightarrow f = \frac{10}{7} mg \Rightarrow \frac{10}{7} mg \leq \mu N$$



$$\Rightarrow \frac{10}{7} mg \leq \mu (5mg), \mu \geq \left(\frac{2}{7} \right)$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (1000)**

$$\frac{n}{2\ell_1} \sqrt{\frac{T}{\rho_1 S_1}} = \frac{n}{2\ell_2} \sqrt{\frac{T}{\rho_2 S_2}}$$

$$\frac{n}{m} = \frac{80}{60} \sqrt{\frac{70.2 \times 1}{2.6 \times 3}}$$

$$\frac{n}{m} = 4$$

$$f = \frac{8 \times 100}{2 \times 80} \sqrt{\frac{936 \times 3}{70.2 \times 10^3 \times 10^{-6}}}$$

$$f = 1000 \text{ Hz}$$

2. **Ans (2)**

$$E_0 = \frac{1}{2} CV_0^2$$

$$E = \frac{1}{2} (2C) V_0^2$$

$$E = 2E_0$$

3. **Ans (5)**

$$P_{\text{initial}} = (-5j + (\cos 30i - 5 \sin 30j) + (-5 \cos 30i + 5 \sin 30j))m$$

$$P_{\text{final}} = (O - (5 \cos 30i + 5 \sin 30j) + \vec{V}_c)m$$

$$P_{\text{initial}} = P_{\text{final}}$$

$$\Rightarrow O = -(5 \cos 30i + 5 \sin 30j) + \vec{V}$$

$$\vec{V}_c = (5 \cos 30i + 5 \sin 30j) = 5 \text{ m/s}$$

4. **Ans (24)**

Bulk modulus is given by

$$B = -\frac{dP}{(dV/V)} \text{ or } \frac{dV}{V} = -\frac{dP}{B}$$

$$\text{or } \frac{P \cdot dV}{B} = -\frac{P dP}{B}$$

$$\frac{P dV}{V} \text{ is the work done per unit volume. The}$$

negative sign implies that a decrease in pressure gives rise to increase in volume and vice-versa.

Hence volume density of elastic potential energy.

5. **Ans (20)**

$$\text{Using } Q = CV$$

6. **Ans (1)**

$$W = \int_0^{\pi/2} f \cdot R d\theta = \frac{\mu mg R \pi}{2} = 1 \text{ joule}$$

7. **Ans (24)**

$$\frac{4m_B m_C g}{m_B + m_C} = m_A g$$

$$\frac{4}{m_A} = \frac{1}{m_B} + \frac{1}{m_C}$$

$$\frac{4}{m} = \frac{1}{2} + \frac{1}{3}$$

$$m = 4.8$$

8. **Ans (8)**

$$\Delta P = 10 - (10) \times 0.2$$

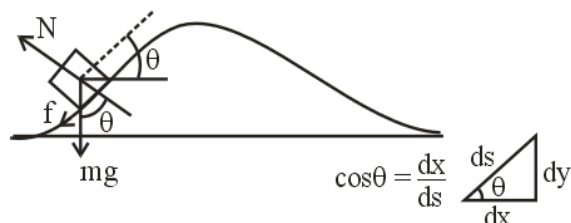
$$\Rightarrow 8 = 1 \times v \Rightarrow v = 8 \text{ m/sec.}$$

9. **Ans (20)**

Work done by friction

$$= \int \vec{F} \cdot d\vec{s} = \int_0^x \mu mg \cos \theta \frac{dx}{\cos \theta}$$

$$= \mu mg x = 20 \text{ J}$$



10. **Ans (5)**

$$\lambda_p N_p dt = \lambda_d N_d dt \Rightarrow \frac{\lambda_p}{\lambda_d} = \frac{N_d}{N_p} = \frac{1}{2}$$

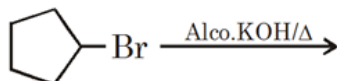
PART-2 : CHEMISTRY

SECTION-I

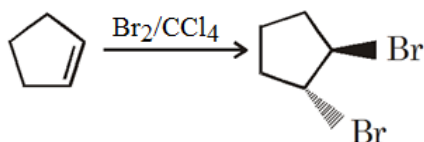
1. **Ans (B)**

NBS carries out bromination at allylic and benzylic carbon.

2. **Ans (D)**

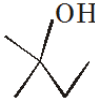


Bromocyclopentane

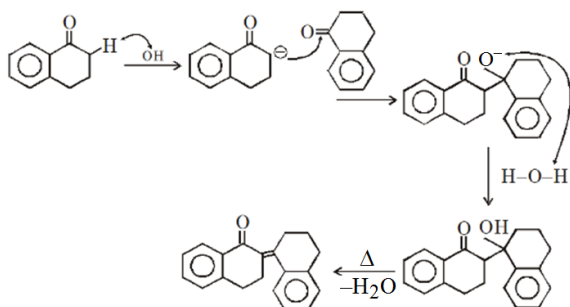


3. **Ans (B)**

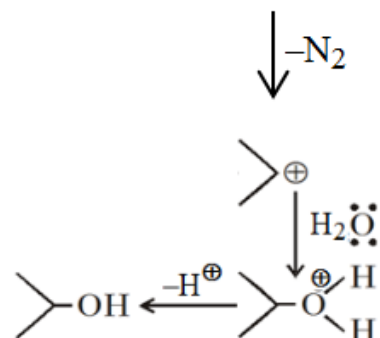
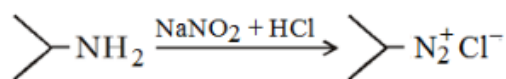
RDS is the formation of carbocation.

Compound  gives most stable carbocation among given compounds.

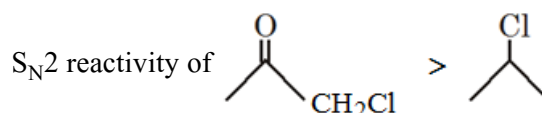
4. **Ans (C)**



5. **Ans (D)**



6. **Ans (C)**

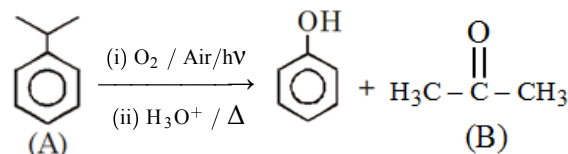


Gutta-percha is Trans-polyisoprene

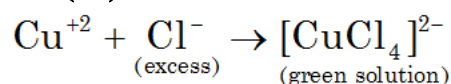
Cellulose contains β -1,4 glycosidic linkages

$\text{CH}_3\text{CH}_2\text{OH}$ gives Haloform test while CH_3OH doesn't

7. **Ans (D)**



8. **Ans (C)**



10. **Ans (A)**

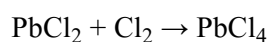
General order of stability of oxides :

I-oxides > Cl-oxides > Br-oxides

11. **Ans (D)**

Mischmetal has : ~95% of lanthanoid + ~5% of Fe and traces of C, S, Ca, Al

12. **Ans (C)**



13. **Ans (D)**

Max. permissible concentration of Cd in drinking water is 0.005 ppm.

15. **Ans (B)**

$$M = \frac{n_2}{V(\text{ml})} \times 1000 \quad \& \quad m = \frac{n_2}{w_1(\text{g})} \times 1000$$

here $V > w_1$ (numerically)

$$\Rightarrow M < m.$$

16. **Ans (C)**

Using $\ln p^\circ = \frac{-\Delta H_{\text{vap.}}}{R} \left(\frac{1}{T} \right) + \text{constant}$ enthalpy of vapourisation : $\Delta H_I > \Delta H_{II}$

17. **Ans (C)**

$$\frac{a}{\sqrt{3}} = 2.31 \times 10^{-8} \text{ cm}$$

$$a \simeq 4 \times 10^{-8} \text{ cm}$$

$$a^3 = 64 \times \frac{10^{-24}}{4 \times 60} \text{ cm}^3$$

$$d = \frac{64 \times 10^{-24}}{64 \times 10^{-24} \times 6 \times 10^{23}}$$

$$d = 6.25 \text{ g/cm}^3$$

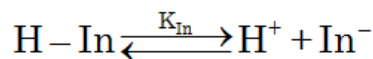
19. **Ans (C)**

$[\text{OH}^-]$ in resultant solution

$$= \frac{50.01 \times 0.01 - 50 \times 0.005 \times 2}{(50 + 50.01)}$$

$$\simeq 10^{-6} \text{ M}$$

$$[\text{H}^+] \simeq \frac{10^{-14}}{10^{-6}} = 10^{-8} \text{ M}$$



$$K_{\text{In}} = \frac{[\text{H}^+][\text{In}^-]}{[\text{H-In}]}$$

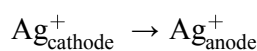
$$2 \times 10^{-8} = 10^{-8} \frac{[\text{In}^-]}{[\text{H-In}]}$$

$$\frac{2}{1} = \frac{[\text{In}^-]}{[\text{H-In}]}$$

$$\text{required fraction} = \frac{[\text{In}^-]}{[\text{In}^-] + [\text{H-In}]} = \frac{2}{3}$$

20. **Ans (D)**

Overall cell reaction can be written as



The cell will be treated as dead cell if

$$[\text{Ag}^+]_{\text{cathode}} = [\text{Ag}^+]_{\text{anode}}$$

$$K_{\text{sp}} \text{ of AgBr}/[\text{Br}^-] = K_{\text{sp}} \text{ of AgCl}/[\text{Cl}^-]$$

$$\frac{[\text{Br}^-]}{[\text{Cl}^-]} = \frac{K_{\text{sp}} \text{ of AgBr}}{K_{\text{sp}} \text{ of AgCl}} = \frac{2 \times 10^{-13}}{8 \times 10^{-10}} = 2.5 \times 10^{-4}$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3)**



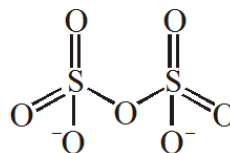
2. **Ans (2)**

Among the given amino acids only histidine and arginine are considered as basic amino acids.

3. **Ans (5)**

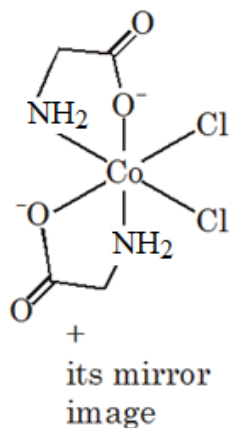
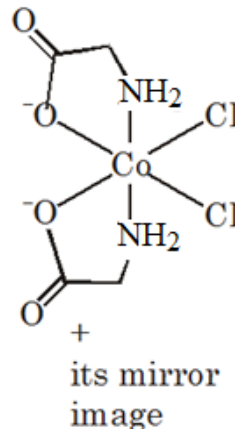
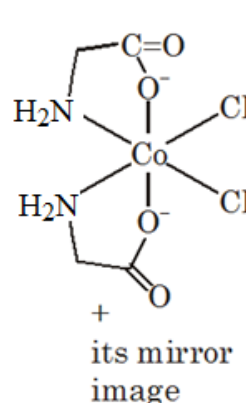
Dacron, Nylon-6, Nylon-66, Bakelite and Novolac are condensation polymers.

4. **Ans (10)**



$$x = \frac{5}{3} \Rightarrow 6x = 10$$

5. **Ans (6)**



6. **Ans (8)**

All except Calamine, Copper matte, Siderite and Pig iron.

7. **Ans (1)**

Case 1 :

$$h\nu = w + (KE)_{\max}$$

$$\frac{hc}{\lambda} = w + KE_1$$

$$\frac{hc}{\lambda} = w + 1\text{eV} \quad \dots(i)$$

Case 2 :

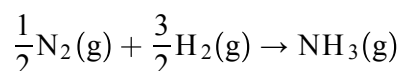
$$h\nu_2 = w + KE_2$$

$$\frac{hc}{\lambda/3} = w + 5\text{eV}$$

$$\therefore \frac{3hc}{\lambda} = w + 5 \quad \dots(ii)$$

On solving both the equations $w = 1\text{ eV}$

8. **Ans (11)**



$$\Delta_f H^\circ = -\frac{1}{2} \times 40 \times 10^3 \text{ cal/mole}$$

$$\Delta_f S^\circ = 1 \times S_{\text{NH}_3}^\circ - \frac{1}{2} S_{\text{N}_2}^\circ - \frac{3}{2} S_{\text{H}_2}^\circ$$

$$= 1 \times 40 - \frac{1}{2} \times 50 - \frac{3}{2} \times 30$$

$$= 40 - 25 - 45$$

$$= -30 \text{ cal/mole/K}$$

$$\Delta_f G^\circ = \Delta_f H^\circ - T \Delta_f S^\circ$$

$$= -20 \times 10^3 + 300 \times 30 = -20,000 + 9000$$

$$= -11 \text{ kcal/mole}$$

9. **Ans (6)**

$$\frac{n_{\text{CH}_4}/t}{n_{\text{He}}/t} = \frac{P_{\text{CH}_4}}{P_{\text{He}}} \sqrt{\frac{M_{\text{He}}}{M_{\text{CH}_4}}}$$

$$= \frac{1}{3} \times \sqrt{\frac{4}{16}} = \frac{1}{6}$$

10. **Ans (3)**

For nth order reaction :

$$r = k[A]^n$$

$$t_{1/2} = \frac{(2^{n-1} - 1)}{(n-1)ka_0^{n-1}} \quad (n \neq 1)$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

Let roots are $\frac{k}{r}, k, kr$

$$\Rightarrow k = -6$$

$$\Rightarrow \frac{k}{r} + k + kr = -a \quad \dots(1)$$

$$\frac{k^2}{r} + k^2r + k^2 = b \quad \dots(2)$$

Solving (1) & (2) we set $\frac{b}{a} = 6$

$$\frac{a^2 + b^2}{ab} \Rightarrow \frac{a}{b} + \frac{b}{a}$$

$$= 6 + \frac{1}{6}$$

$$= \frac{37}{6}$$

2. **Ans (C)**

Put $5^x = t$

$$t^2 + t - m < 0$$

For at least one $t \in \mathbb{R}^+$

$$\text{Since } -\frac{b}{2a} < 0$$

$$\Rightarrow f(0) < 0$$

$$-m < 0$$

$$\Rightarrow m > 0$$

3. **Ans (C)**

Distance from origin remains same

$$\frac{1}{4^2} + \frac{1}{5^2} = \frac{1}{8^2} + \frac{1}{k^2}$$

$$\Rightarrow k = \pm \frac{40}{\sqrt{139}}$$

$$|k| = \frac{40}{\sqrt{139}}$$

4. **Ans (D)**

$$AD = 4$$

$$AP = \frac{2}{3} \times 4 = \frac{8}{3}$$

In $\triangle ABP$

$$\frac{BP}{AP} = \tan \frac{\pi}{6} \Rightarrow BP = \frac{8}{3\sqrt{3}}$$

$$\text{Area of } \triangle ABC = 3 \times \text{Area of } \triangle ABP = \frac{32}{3\sqrt{3}}$$

5. Ans (A)

$g(x) = 12x^5 + 15ax^4 + 20bx^3 + 30cx^2 + 60dx + k = 0$
has 5 positive real roots.

∴ By Rolle's theorem $g'(x) = 0$ has four positive real roots.

∴ $f(x) = 0$ has four positive real roots $\alpha, \beta, \gamma, \delta$ such that
$$\frac{\alpha + \beta + \gamma + \delta}{4} = \frac{4}{(1/\alpha) + (1/\beta) + (1/\gamma) + (1/\delta)}$$

$$\Rightarrow AM = HM \Rightarrow \alpha = \beta = \gamma = \delta$$

$$\therefore f(x) = (x - \alpha)^4 \Rightarrow f(2) = (2 - \alpha)^4 = 1$$

$$\Rightarrow 2 - \alpha = 1 \text{ or } \alpha = 1 \text{ or } 3.$$

6. Ans (C)

$$\text{Put } \int_0^{\pi/2} \theta \sin^8 \theta d\theta = I, \int_0^{\pi/2} \theta \sin^{10} \theta d\theta = P$$

$$P = \int_0^{\pi/2} (1 - \cos^2 \theta) \sin^8 \theta d\theta = I, - \int_0^{\pi/2} \theta \cos^2 \theta (\sin^8 \theta) d\theta$$

$$= I - \int_0^{\pi/2} (\theta \cos \theta) (\cos \theta \sin^8 \theta) d\theta. \text{ Using integration}$$

by parts we get

$$P = I - \left[\frac{(\theta \cos \theta) \sin^9 \theta}{9} - \int \left(\frac{-\theta \sin \theta + \cos \theta}{9} \right) \sin^9 \theta d\theta \right]_0^{\pi/2}$$

$$P = I - \left(\frac{P}{9} \right) + \frac{1}{9} \int_0^{\pi/2} \sin^9 \theta \cos \theta d\theta, \frac{10P}{9} = I + \frac{1}{90}$$

$$\therefore 100P = 90I + 1$$

$$\therefore \frac{100P - 1}{I} = 90$$

7. Ans (D)

By inspection $f(\pi/2) = 3\pi$, as $g(x) = f^{-1}(x)$, $g(f(x)) = x$.

Taking derivative we get $f'(x) g'(f(x)) = 1$ (i)

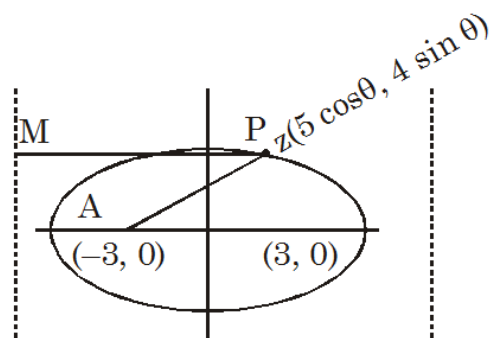
Taking derivative again we get $f''(x) g'(f(x)) + f'(x)^2 g''(f(x)) = 0$ (ii)

$$f(x) = 4x + \pi \sin x \text{(iii)}$$

From (iii) we can get the value of $f'(\pi/2)$ and $f''(\pi/2)$.

By substituting $x = \pi/2$ in (i) and (ii) we can get the value of $g'(3\pi)$ and $g''(3\pi)$.

8. Ans (A)



$$2ae = 6, 2a = 10$$

$$\Rightarrow a = 5, b = 4, e = \frac{3}{5}$$

$$\frac{60 |z + 3|}{|z + \bar{z} + (50/3)|} = \frac{60 \times PA}{10 \cos \theta + (50/3)}$$

$$= \frac{60 \times e \times PM}{10 \cos \theta + (50/3)} = \frac{60 \times \frac{3}{5} \times (5 \cos \theta + \frac{25}{3})}{10 \cos \theta + (50/3)} = 18$$

9. Ans (D)

$$\sqrt{2} (\cos 3x - \sin 3x) = \sqrt{3} \sin 2x + \cos 2x$$

$$\frac{(\cos 3x - \sin 3x)}{\sqrt{2}} = \left(\frac{\sqrt{3} \sin 2x + \cos 2x}{2} \right)$$

$$\cos \left(3x + \frac{\pi}{4} \right) = \cos \left(2x - \frac{\pi}{3} \right)$$

$$3x + \frac{\pi}{4} = 2n\pi \pm \left(2x - \frac{\pi}{3} \right)$$

$$x = 2n\pi - \frac{7\pi}{12}, x = \frac{2n\pi}{5} + \frac{\pi}{60}; n \in \mathbb{Z}$$

10. Ans (A)

$$\text{Differentiate to get } y' = \frac{-9(x - \sqrt{1+x^2})^9}{\sqrt{1+x^2}}$$

$$\text{Rewrite as } \sqrt{1+x^2} \cdot y' = -9y \text{(1)}$$

Differentiate (1) both sides w.r.t. x , then put value of $\sqrt{1+x^2} \cdot y'$ from (1).

11. Ans (D)

$$1 - (1/4)^n \geq 8/9 \Leftrightarrow 4^n \geq 9 \Leftrightarrow n \log 4 \geq 2 \log 3$$

12. Ans (C)

Let ED be the pole and EC the flagstaff. A be the nearer point. $AD = DC = 4 \Rightarrow \angle CAD = 45^\circ$
so solve $\triangle EAD$.

13. Ans (C)

Point of intersection is (2, 1). Slope of tangents are $1/2$ and -2 respectively.

14. **Ans (B)**

Domain $-1 \leq x \leq 1$. From graph of functions $y = \cot^{-1} x$ and $y = \sin^{-1} x$, solution must be of the form $[-1, \alpha)$ where $0 < \alpha < 1$ satisfies $\cot^{-1} \alpha = \sin^{-1} \alpha$. Take sin on both sides, put $\cot^{-1} \alpha = \theta$ and solve using right angled triangle.

15. **Ans (C)**

p means 'at least one prime number is even'.
So its negation is 'No prime number is even'.

16. **Ans (A)**

Let x be thickness of ice at time t.

$$\text{So } \frac{d}{dt} \left(\frac{4}{3} \pi (20+x)^3 - \frac{4}{3} \pi (20^3) \right) = -50.$$

Use chain rule and put $x = 5$.

17. **Ans (D)**

Domain is R. Clearly we need to check differentiability only at $x = 0$. Use first principle.

18. **Ans (C)**

Centre $(2, k)$ and passing through $(0, 0)$ so

$$(x-2)^2 + (y-k)^2 = 2^2 + k^2$$

$\Rightarrow x^2 + y^2 - 4x - 2ky = 0 \dots (1)$ differentiate and eliminate k.

19. **Ans (C)**

H : hit, X : not hit. $P(HXXH \text{ or } XHXX \text{ or } XXHH) = 3 \times (2/3)^2 \times (1/3)^2$.

20. **Ans (C)**

Put $x - \pi/4 = t$, expand numerator and integrate.
OR check options by differentiating.

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (6)**

Solve first two equations to get $x = 2 + z$ and $y = 4 - 2z$. Put in third equation to get $(\lambda - 6)z = 4$. No solution for $\lambda = 6$.

$$\text{OR } \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 4 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 6.$$

2. **Ans (0)**

$$2^{(-k+3^{-a})} = 2^{-k} \cdot 2^{3^{-a}}$$

$$\text{where } 3^{-a} = \frac{1}{\log_3 4} = \frac{1}{2 \log_3 2} = \frac{\log_2 3}{2}.$$

$$\text{So } 2^{3^{-a}} = (2^{\log_2 3})^{1/2} = \sqrt{3}$$

3. **Ans (0)**

$A^2B + AB^2 = 0$. Pre-multiply both sides by A to get

$$A^3B + A^2B^2 = (-B^3)B + A^2B^2 = (-B^2 + A^2)B^2 = 0.$$

Take determinant on both sides to get $|A^2 - B^2| = 0$

or $|B| = 0$. If $|B| = 0$ then taking determinant on both

sides of $A^3 = -B^3$ gives $|A| = 0$, which is contradiction.

4. **Ans (1)**

Multiply and divide denominator by x^2 to remove $\sin^2 x$ then use L' Hospital rule (Use Leibniz rule for differentiation of integral)

5. **Ans (20)**

Let number of elements in S be k. $10k = 5n$ and $9k = 90$.

6. **Ans (4)**

$AB = I$. So $(1, 1)^{\text{th}}$ element of AB must be 1.

7. **Ans (624)**

$2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$ Required sum is $(3^0 + 3^1 + 3^2)(5^0 + 5^1)(7^0 + 7^1)$.

8. **Ans (11)**

Put $z = \omega$ or $z = \omega^2$.

9. **Ans (2)**

Direct method : area = $\int_{(1-e)/2}^0 \log_e (2x + e) dx$.

Now use integration by parts. Faster method : area

$$= \int_0^1 x dy = -(1/2) \int_0^1 (e^y - e) dy$$

10. **Ans (2)**

Replace x by $a + b - x$ then add the two integrals